

From Historical Mobility Orchestration to Minimum Sufficient Control

Case note on system externalities, cascading commitments and Pareto-based scenario planning in the Riviera Maya

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Purpose and proposed contribution

Tegrity.AI is developing Minimum Sufficient Control (MSC) as a generalizable mobility-systems architecture: how much coordinated intervention is actually required to keep declared mobility outcomes within an acceptable objective envelope, without assuming that effective policy requires system-wide direct control. The work examines three linked questions: how a bounded managed fleet interacts through network interdependencies; how committed demand and interdependent capacity or timing obligations create cascades; and how Pareto-optimal scenario alternatives shape accountable multi-objective decisions.

For WP.5, the proposed contribution is to develop policy-relevant methods, comparable measures and cross-case questions for three recurring sustainable-mobility challenges: when partial coordination affects network interdependencies; how mobility architectures can remain resilient when local disturbances propagate; and whether transparent multi-objective scenarios and accountable human choice improve decisions compared with a consolidated utility function.

Historical operational substrate: Riviera Maya, 2016–2017

From 2016, a tourism operator used a passenger-mobility orchestration system in the Riviera Maya to collect passengers from approximately 400 hotels and serve seven destination parks. Instead of fixed itineraries, the system generated constrained route sheets for each bus and dynamically replanned assignments, routes, timing, access and occupancy across hundreds of vehicles, coordinating between 7,000–15,000 passenger movements per day [3]. Because those buses used shared corridors and access points, urban traffic forecasts were an operating input to the plan.

The preserved records make it possible to compare planned assignments, actual vehicle movements and wider traffic conditions across the same corridors and time bands.

Three phenomena developed from the case

Phenomenon	What the case makes observable	Policy / analytical question
1 • System externalities	A bounded managed fleet concentrates movements in shared corridors and access points; network interdependencies affect travel times and plan feasibility.	How should policy measure positive or negative system-level effects when direct coordination covers only part of total mobility?
2 • Cascading commitments	A local change in demand, vehicle availability, timing or capacity can invalidate an assignment and propagate through later groups, routes and transfers.	How should reliability and resilience represent cascade depth, lost slack, repair scope and criticality thresholds for wider replanning?
3 • Pareto scenario planning	The preserved planner searched Pareto-optimal alternatives across explicit objectives and exposed scenario-level scores before operator selection.	Can urban mobility choices be made more transparent through executable multi-objective alternatives rather than one opaque “optimal” plan?

1 • System externalities: managed-fleet decisions and shared-network outcomes

The empirical opportunity is to align planned concentrations, routes and timing with actual vehicle movement and independent corridor/time-band traffic conditions. The analysis can examine whether coordinated fleet movements coincided with changes in congestion, travel-time distributions or flow conditions, and whether those external conditions fed back into the managed operation. Stern et al. and the later CIRCLES field programme provide related evidence that limited direct intervention and system-level effects need not be one-to-one. The policy question is how to measure positive or negative network effects and interdependencies when coordination covers only a bounded share of mobility.

2 • Cascades: when a local change stops being local

Under fully committed demand, each accepted assignment consumes seats, time and route flexibility. A late reservation, cancellation, vehicle loss or timing disruption can therefore alter the feasible set for subsequent commitments. The preserved records allow the work to identify critical failures and bifurcation points: the moments

when the conditions on which a decision was based no longer describe the current system, local repair ceases to be sufficient and the architecture must reconfigure or replan. This supports a policy-relevant question that extends to smart-mobility decisions generally: how should mobility architectures contain cascades and remain resilient as commitments, constraints and system state change? In policy terms, this also raises a criticality threshold: when does a local disturbance cease to be locally absorbable and require wider reconfiguration? Related branching and optimization questions are also under exploratory discussion with QAVA at the Universitat de València; no independent technical validation is claimed.

3 • Pareto scenarios: transparent trade-offs before execution

The historical planner searched Pareto-optimal alternatives across explicit priorities and exposed scenario outcomes before commitment. Historical objectives included punctuality/SLA, policy compliance, occupancy and vehicle use, directness/travel time and fuel-use reduction.

The retained evidence can be used to ask whether presenting Pareto-optimal scenarios helped human controllers make better decisions than a consolidated utility function would have done. The analysis would examine positive, negative or neutral effects of this form of white-box decision support and human-in-the-loop selection after feasible Pareto alternatives had been generated. This provides an empirical basis for examining transparency, accountability and decision quality. It may also provide a short-horizon operational counterpart to scenario-based policy stress-testing: alternative executable plans can be compared before commitment rather than only after implementation.

Integration: Minimum Sufficient Control

MSC integrates the three axes. “Minimum” is not a fixed percentage of vehicles, and “control” does not imply maximum sensing or direct command of the entire mobility population. A coordination arrangement is sufficient only if it can keep declared outcomes inside an acceptable objective envelope while absorbing relevant cascades and accounting for material network interdependencies with the shared system. The object of study is the least intervention-intensive arrangement that satisfies those conditions.

Externality	Cascade	Objective envelope
Account for outward and inward effects between the coordinated subset and the shared network.	Test whether disturbances remain bounded or propagate beyond the intended coordination scope.	Compare explicit multi-objective scenarios and define which outcome combinations are acceptable.

Possible alignment with WP.5

Tegrity.AI proposes to develop these three dimensions as a possible input to WP.5's sustainable urban mobility work. The analysis can produce a reusable case-assessment template, concise methods, comparable measures and cross-case questions applicable to public or private managed fleets across cities with different institutional, technological and resource capacities.

A practical alignment could focus on three outputs:

Managed-mobility case assessment template	A compact structure covering coordination scope, network interdependencies, cascade/criticality thresholds and scenario/objective envelope.
Comparable measures	Candidate fields such as external-effect windows, cascade depth / repair scope, criticality thresholds, slack loss and scenario-level objective scores.
Cross-case learning	A structure reusable across operators and cities without assuming identical technology, governance or capacity.

MSC remains a working architecture rather than a closed formulation. The Riviera Maya evidence is being used to test which variables are genuinely necessary, which can be measured reliably, and which distinctions remain useful when moving from one operator or city context to another.

Tegrity.AI would welcome Secretariat guidance on whether this analytical input would be useful to present or future WP.5 exchanges, workshops or recommendations. WP.5 could also identify questions it would like the historical evidence to address—for example, whether the operation embodied particular architectural properties, coordination mechanisms or resilience criteria—and Tegrity.AI could examine the preserved case accordingly.

References and evidence basis

[1] UNECE, Working Party on Transport Trends and Economics (WP.5), 39th session, 23–25 September 2026; annotated provisional agenda ECE/TRANS/WP.5/79.

[2] UNECE, ECE/TRANS/WP.5/2025/6, WP.5 evaluation report and updated programme of work, including sustainable urban mobility activities for 2026–2027.

[3] Contributor-retained xSeil evidence archive (2016–2017): preserved source code, planning configurations, bookings, route sheets, execution records, vehicle/geolocation data and operational reports.

[4] R. E. Stern et al., “Dissipation of stop-and-go waves via control of autonomous vehicles: field experiments,” *Transportation Research Part C* 89 (2018), 205–221.

[5] M. Ameli et al., “Design, Preparation, and Execution of the 100-AV Field Test for the CIRCLES Consortium: Methodology and Implementation of the Largest Mobile Traffic Control Experiment to Date,” *IEEE Control Systems* 45(1) (2025), 139–155. <https://doi.org/10.1109/MCS.2024.3500332>.